REAR-IR1: rear docking IR board, 1 3.3 V, 2 IR_L (PC8, TIM3_CH3), 3 IR_R (PC9, TIM3_CH4), 4 GND.
HALL-BIN1 / HALL-TOWER1: Hall boards, SH 3-pin (ZX-SH1.0-3PLT, so a Hall cable cannot go into a 4-pin socket): 1 3.3 V, 2 OUT (PE4 / PE5, low = magnet near), 3 GND. All mounting pads to GND.

CLIFF
CLIFF-LMD
CLIFF-RMD
CLIFF_LED_EN
CLIFF-RRD
CLIFF-RFD
CLIFF-LRD
CLIFF-LFD
File: CLIFF.kicad_sch

MAIN BRUSH MOTOR
MAIN-BRUSH-IN1
MAIN-BRUSH-IN2
MAIN-BRUSH-ADC
File: MAIN-BRUSH-MOTOR .kicad_sch

WHEEL MOTORs
WHEEL-MOT-R-IN2
WHEEL-MOT-R-IN1
WHEEL-MOT-R-ENCODED
WHEEL-MOT-R-DROPD
WHEEL-MOT-L-IN2
WHEEL-MOT-L-IN1
WHEEL-MOT-L-DROPD
WHEEL-MOT-L-ENCODED
WHEEL-MOT-R-ADCD
WHEEL-MOT-L-ADCD
File: WHEEL-MOTORs .kicad_sch

MAIN FAN
MAIN-FAN-FBD
MAIN-FAN-PWM
File: MAIN-FAN .kicad_sch

SIDE BRUSH MOTORs
SIDE-BRUSH-IN1
SIDE-BRUSH-IN2
SIDE-BRUSH-ADCD
File: SIDE-BRUSH-MOTORs .kicad_sch

Bumper Sensors
BUMPER
BUMP-L
BUMP-R
BUMP_L_LED_EN
BUMP_R_LED_EN
File: bumper_sensors.kicad_sch

VSENS-3V3
SIDE-R1
ZX-SH1_0-7PWT

USART2-IR-R-RX
SIDE-PROXI-CE-R
I2C4_SCL
I2C4_SDA

MCU-STM32
CLIFF-LF
CLIFF-LR
CLIFF-RF
CLIFF-RR
CLIFF_LED_END
CLIFF-RM
CLIFF-LM
BUMP-L
BUMP-R
BUMP_L_LED_END
BUMP_R_LED_END
USART2-IR-R-RX
SIDE-PROXI-CE-RD
I2C4_SDA
I2C4_SCLD
CARPET-SENSOR-LOD
CARPET-SENSOR-ECHO
WDID
LIDAR_RXD
LIDAR_TX
LIDAR-MOTOR-CTRLD
INT-SENSORS
VL-L-LPnD
VL-I2C-RESETD
UART3_RX2
UART3_RX1
DP
DN
USB_CC1D
USB_CC2
USB_VBUS_SENSE
USB_PWR_END
VMOT-14V4-END
REAR-IR-L
REAR-IR-R
IMU-INT#1
LED-HOME
LED-PWR
CLEAN-BTN
HOME-BTN
BAT_ID
DC_IN_SENSE
BAT_SENSE
VSENS_5V_USB_EN
VSENS_3V3_EN
PWR-BTN-SENSE
STM-PWR-CTRL
HALL-BIN
HALL-TOWER
File: STM32G473.kicad_sch

VSENS-3V3
R61
47k
INT-SENSORS
VL-L-LPn
VL-I2C-RESET

CARPET-SENSOR
LO
ECHO_ADCD
File: Carpet-sensor.kicad_sch

WATCHDOG
WD
WDI
WD_OKD
File: WATCHDOG.kicad_sch

WD_OK

LiDAR
WD_OK
LiDAR-RXD
LiDAR-TXD
LIDAR-MOTOR-CTRL
LiDAR-EN
File: LIDAR.kicad_sch

FRONT SENSORS
INT-SENSORS
VL-L-LPn
VL-I2C-RESET
IR-F-RX2
IR-F-RX1
VL-I2C-SCL
VL-I2C-SDA
File: Front-Sensors.kicad_sch

USB Power
USB-C POWER
DP
DN
USB_CC1
USB_CC2
USB_VBUS_SENSE
USB_PWR_EN
File: usb_power.kicad_sch

HALL-BIN1 VSENS-3V3
ZX-SH1_0-3PLT
HALL-TOWER1 VSENS-3V3
ZX-SH1_0-3PLT
REAR-IR1 VSENS-3V3
REAR-IR-L
REAR-IR-R
ZX-SH1_0-4PLT

VSENS-3V3
R6 2.2k
R5 2.2k

IMU
IMU-INT#1
IMU-SDA
IMU-SCL
File: IMU-ICM-4267-P .kicad_sch

POWER CONTROL
VMOT-14V4-EN
STM-PWR-CTRL
PWR-BTN-SENSE
VSENS_3V3_EN
VSENS_5V_USB_EN
WD_OK
PWR-SW-G
POWER-BTN
File: SYSTEM-POWER.kicad_sch

BUTTON-LEDs
POWER-BTND
LED-HOME
LED-PWR
CLEAN-BTND
HOME-BTND
File: BUTTON-LEDs .kicad_sch

BATTERY CHARGER
BATTERY-CHARGER
PWR-SW-G
BAT_SENSE
DC_IN_SENSE
ID
File: Battery-Charger.kicad_sch

Sheet: /
File: Main.kicad_sch
Title:
Size: A3 | Date: | Rev:
KiCad E.D.A. 10.0.4 | Id: 1/17

5V USB, SENSORS, LiDAR POWER RAIL
– set 5V_EN HIGH to enable

TPS54560B (U21, U22):
– non–synchronous
– COMP 4.7k + 47nF (all–ceramic Cout, no pole cap)
– SS56 catch diode; Fsw 418 kHz (RT 232k)

VBAT–14V4
PWR_FLAG
C119 100nF/50V
10 R146
U22
TPS54560BDDAR
BOOT
VIN
EN
RT/CLK
PAD
SW
GND
COMP
FB
L3
CYA0630–3_3UH
D43
SS56
VSENS–5V–USB
D46
YLED0603B
VSENS_5V_USB_END
C115 10uF/50V
C116 10uF/50V
C117 100nF/50V
R206 232k
R168 100k
R147 4.7k
C120 47nF/50V
R148 12k
R149 2.2k
C121 10uF/10V
C122 10uF/10V
C147 10uF/10V
R89 1k

VBAT–14V4
Q1 AO4407C
PWR_FLAG
VMOT–14V4
R165 10k
R36 10
C62 10uF/25V
D37 SS34
VMOT–SW–G
Q37 AO4407C
R222 10
VMCU–3V3
R184 10k
Q19 AO3400
WD_OK
VMOT–14V4–EN

MOTOR POWER RAIL 14.4V

– on/off motor power, no soft ramp
– set VMOT–14V4–EN LOW to enable power
– floating (MCU flash/reset) disables power
– WD_OK LOW shuts power off

VMOT–14V4: Q1 + Q37 AO4407C in parallel
– ~7.5 A normal (with the MSD–G fan) is ~0.25 W each
– short stall peaks (~13 A) fine with copper under the drain
– Vgs +/–25 V, no gate divider needed

VMCU–3V3
Q16 AO3401
VSENS–3V3
PWR_FLAG
R90 100k
C176 1uF/25V
D45 YLED0603B
R98 1k
VSENS_3V3_END
R195 10k
C64 100nF/50V

3.3V SENSORS POWER RAIL

– C176 keeps Q16 off when power switches ON
– C64 soft–ramps power
– set VSENS_3V3_EN LOW to enable sensors power

VBAT–14V4
Q13 AO4407C
POWER–EN
R67 10k
C14 1uF/25V
R171 100k
R16 100k
PWR–SW–G
POWER–BTN
C22 1uF/25V
VMCU–3V3
R173 10k
PWR–BTN–SENSE
D26 1N4148
R172 10k
R190 1k
Q23 AO3400
STM–PWR–CTRLD

STM–PWR–CTRL need to use for both EN and OFF from STM32 GPIO pin.
1– Hold EN to HIGH after buttone pressed
2– OFF board

PWR–LATCH
VMCU–3V3
R193 1M
R191 100k
R192 47k
Q29 AO3400
C172 10uF/10V
U23 74LVC1G17
C173 100nF/50V
Q30 AO3400

Hardware off timer: holding the power button for ~4–9 s turns the MCU rail off even if the firmware hangs (Q29 releases C172, U23 trips, Q30 pulls PWR–LATCH low). Firmware turns off at 3 s.
While charging, the rail–hold (Q31 on BATTERY–CHARGER, from DC–IN–20V) keeps the MCU rail on regardless, and charger power alone turns the MCU rail on (Q31 pulls PWR–SW–G low).

3.3V MCU POWER RAIL

– POWER_EN is 14.4V
– set POWER_EN HIGH to enable

TPS54560B (U21, U22):
– non–synchronous
– COMP 4.7k + 47nF (all–ceramic Cout, no pole cap)
– SS56 catch diode; fsw 418 kHz (RT 232k)
– EN abs max 8.4 V: R207/R208 on U21

VBAT–14V4
C111 100nF/50V
10 R141
U21
TPS54560BDDAR
R207 100k
R208 22.1k
L2
CYA0630–3_3UH
D42 SS56
PWR_FLAG
VMCU–3V3
R205 232k
R143 4.7k
C112 47nF/50V
R144 69.8k
R145 22.1k
C106 10uF/50V
C95 10uF/50V
C102 100nF/50V
C113 10uF/10V
C114 10uF/10V
C146 10uF/10V
D44 YLED0603B
R14 1k

14.5V ULTRASONIC CARPET SENSOR POWER

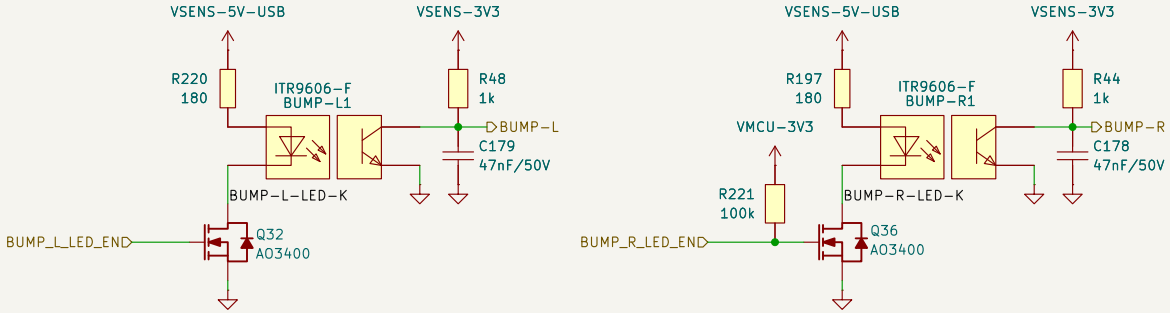
L4
XRNR4030–10UH_M
D4
SS34_C8678
PWR_FLAG
VUSONIC–14V4
VSENS–5V–USB
U5
NC
IN
EN
SW
GND
FB
MT3608
R160 232k
R159 10k
C149 10uF/10V
C148 10uF/10V
C150 10uF/25V
C151 10uF/25V

VUSONIC–14V4 = 0.6V x (1 + 232k/10k) ~= 14.5V

VBAT–14V4
MCU VBAT
– Low quiescent current 2.5µA
– Zener diode shuts off when main battery is low
D2 ZMM9V1_C2891750
PWR_FLAG
U16
HT7533–1_C14289
Vout
Vin
GND
VMCU–BAT–3V3
C140 10uF/25V
C141 10uF/25V

MP1 MP2 MP3 MP4 MP5 MP6

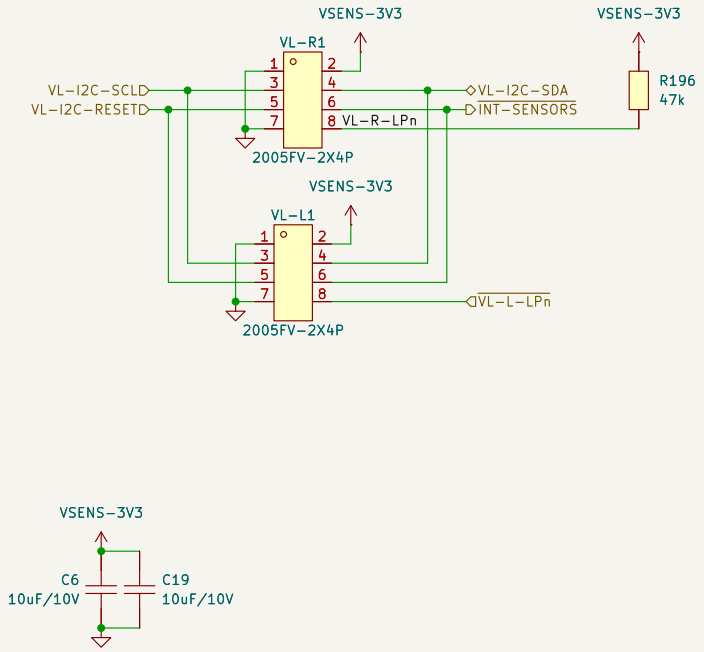
Sheet: /POWER/
File: SYSTEM–POWER.kicad_sch
Title:
Size: A3 | Date: | Rev:
KiCad E.D.A. 10.0.4 | Id: 2/17



- slotted photo-interrupters
- BUMP-X reads low when light passes the slot
- drive BUMP_L/R_LED_EN PWM <<15 usec period to enable
- tune R44, R48, LED PWM duty to let ADC sense full open/close range
  1k keeps photo transistor output out of saturation
  so PA4/PA5 (ADC2) see the transition (ITR9606 IC(ON) 0.5-10 mA)
- R221 enables MCU USB-C Rd on both CC1 and CC2 during MCU *reset*,
  so the phone detects the robot whichever way the cable is plugged in

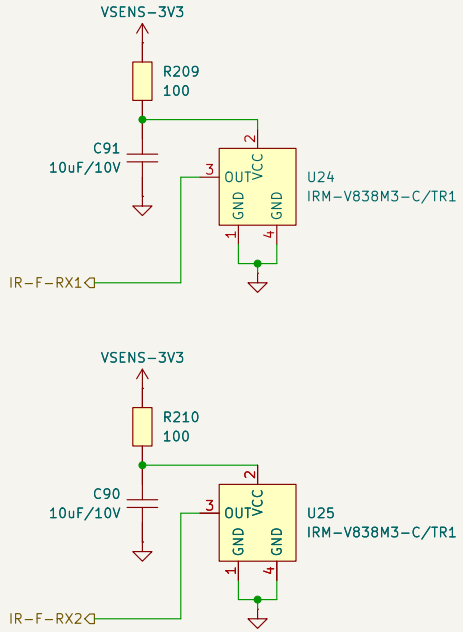
Sheet: /Bumper Sensors/ File: bumper_sensors.kicad_sch		
Title:		
Size: USLetter	Date:	**Rev:**
KiCad E.D.A. 10.0.4		Id: 3/17

2x VL53L7 ToF BOARDS

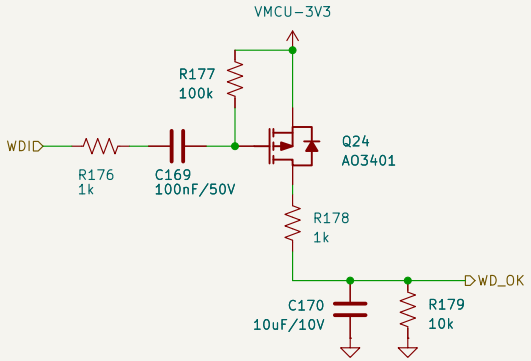


2 IR RECEIVERS

U24 left, U25 right (robot view). Side-looking: put both at the board front edge, lenses facing forward, ~10 mm apart, with a baffle between them in the bumper window. – 90 deg FoV



Sheet: /FRONT SENSORS/ File: Front-Sensors.kicad_sch		
Title:		
Size: USLetter	Date:	**Rev:**
KiCad E.D.A. 10.0.4		Id: 4/17



RC CHARGE-PUMP WATCHDOG

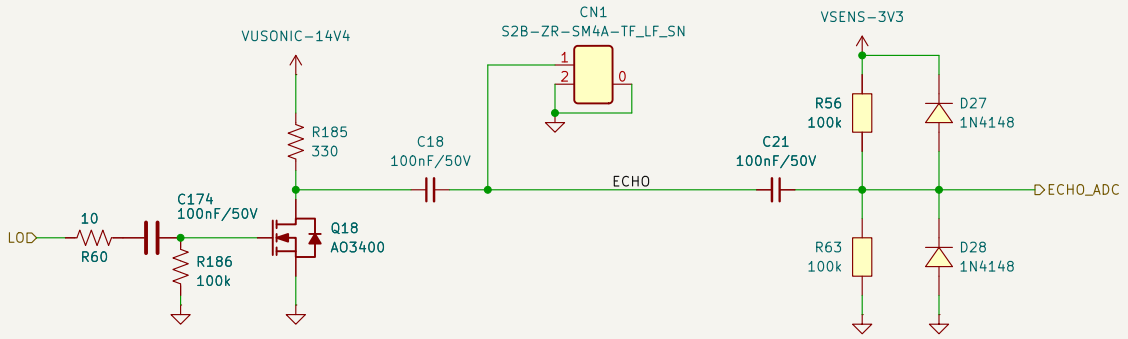
- WD_OK active HIGH
- WDI HIGH->LOW transition pets the dog

Firmware toggles WDI in software (50-1000 Hz, ~50% duty) only while the control loop is healthy.
Never drive WDI from a timer/PWM: a timer keeps running after the CPU hangs.
Each falling edge on WDI pulls the Q24 gate low through C169 and tops up C170 via R178.
WD_OK ~2.7 V while toggling. WDI stuck high, low or floating: <1 V in ~100 ms, <0.4 V in ~190 ms
(10 uF nominal; DC-bias derating shortens this). WD_OK starts at 0 V at power-up (fail-safe).
WD_OK drives N-FET gates only (14.4 V rail switch, LiDAR incl. laser): no resistive loads.
WDI pin: high-Z at reset, no pull-down.

Sheet: /WATCHDOG/	
File: WATCHDOG.kicad_sch	
Title:	
Size: USLetter	Date:
KiCad E.D.A. 10.0.4	Id: 5/17

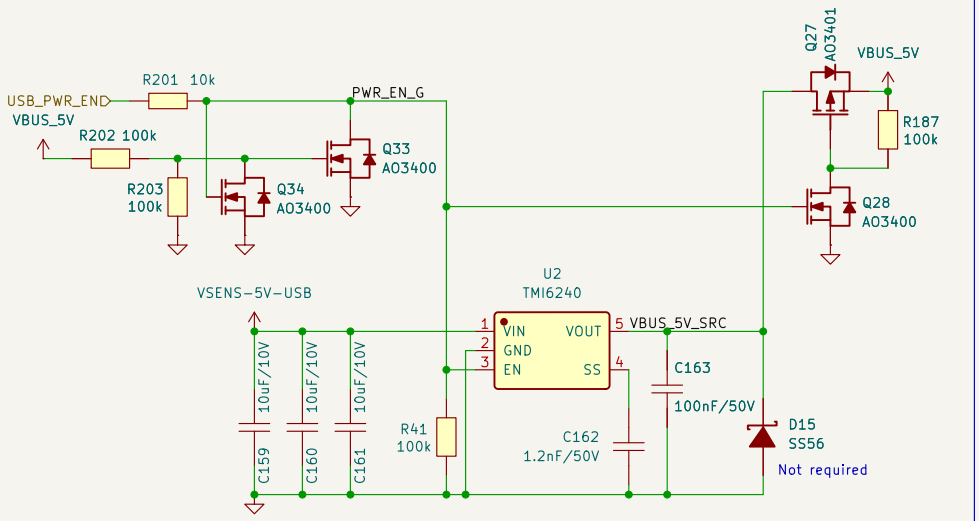
Carpet sensor driver: one low-side AO3400 + 330R pull-up from VUSONIC-14V4 (14.4 V). The P-FET half-bridge was removed: its 100k gate pull-up could not turn it off at 290 kHz (shoot-through). R186 holds LO low in reset/flash.
Drive LO in short bursts only: LO stuck high puts 0.63 W into R185.

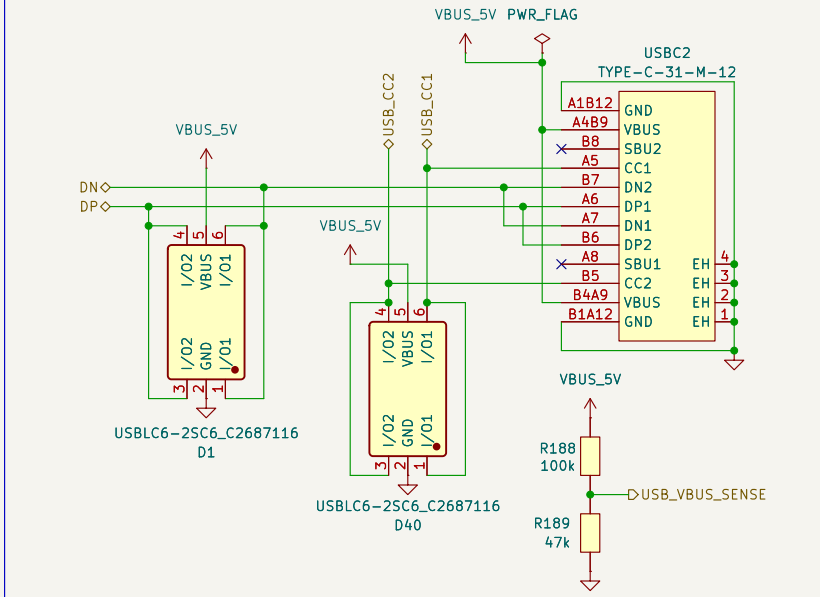
C174 AC-couples LO to the Q18 gate: if LO sticks high, the gate decays through R186 (~10 ms) and Q18 turns off, so R185 is never stressed by firmware holding LO high.



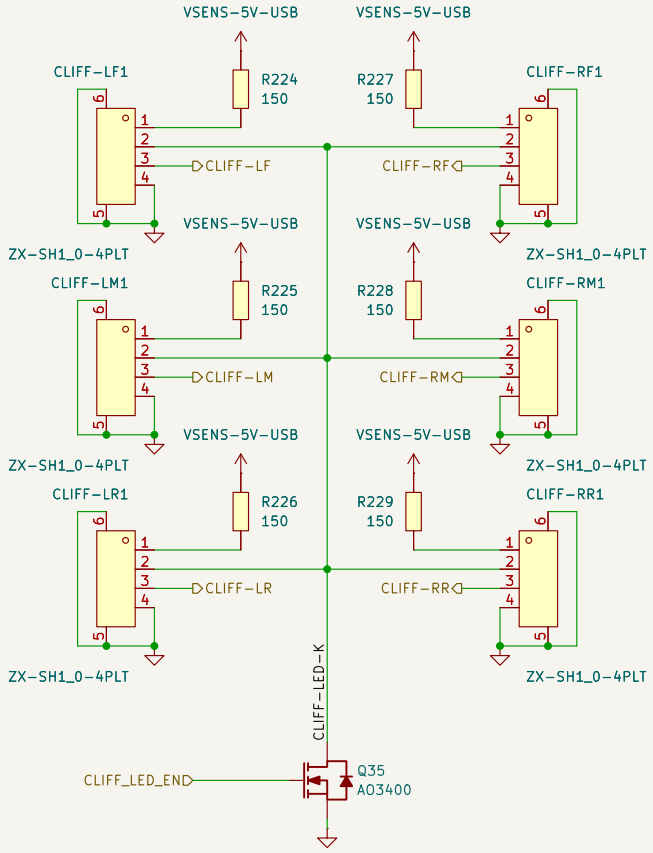
Sheet: /CARPET-SENSOR/		
File: Carpet-sensor.kicad_sch		
Title:		
Size: USLetter	Date:	**Rev:**
KiCad E.D.A. 10.0.4		Id: 6/17

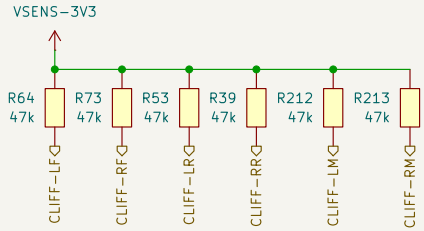
USB-C for SMARTPHONE





Sheet: /USB Power/	
File: usb_power.kicad_sch	
Title:	
Size: USLetter | Date:	**Rev:**
KiCad E.D.A. 10.0.4	Id: 7/17



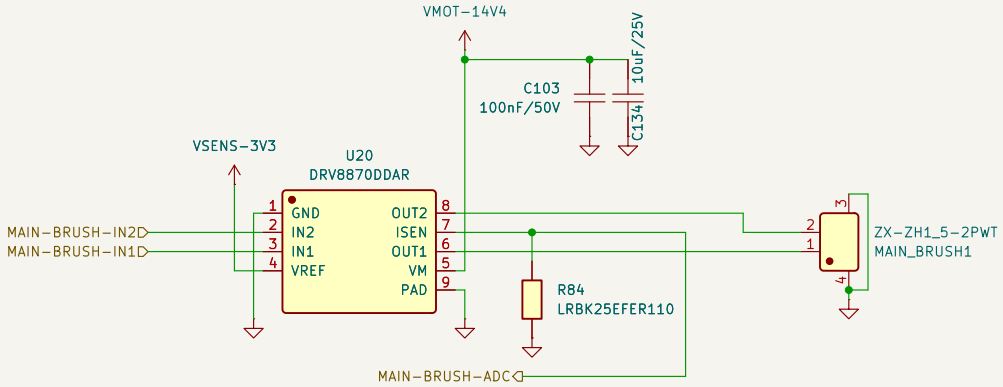


- kicad/cliff-sensor mating boards
- SH 4-pin vertical; use same (pin 1 to pin 1) cable
- set CLIFF_LED_EN LOW to disable
- modulate CLIFF_LED_EN, read each channel LED-on minus LED-off to cancel ambient light
- possible TODO - cliff LEDs 2x or 3x in series, save resistors
  - worsens power rail noise rejection
  - ADC fw could reject noise by correlating the 6 readings
- 0.76W power consumption when enabled continuously
  - drive at ~10% duty cycle to save power
- tweak output pullup
  - a low resistor value can saturating the photo transistor
  - 47K pullup makes for 1ms output settling time
  - can use 10k for faster settling, to reduce duty cycle further
- combine low duty with fast duty LED cycling
  - fast (~10x photo transistor settling time) reduces LED average radiance to avoid photo transistor saturation
- check sensor faults - is output 3.3V or floating?
  - normal no-cliff condition - photo transisstor gets LED light Vary LED current and check photo-transistor voltage.

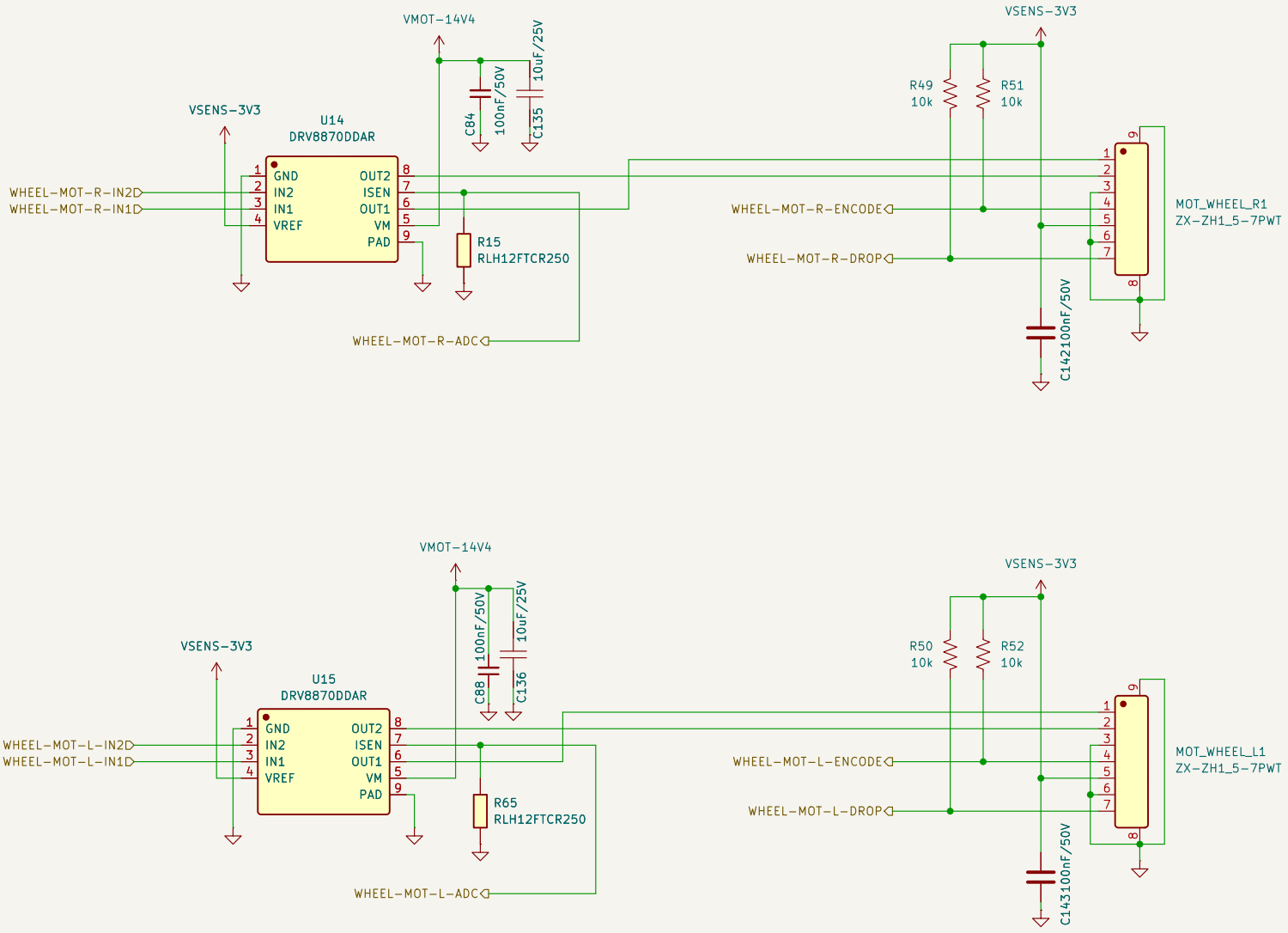
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Title:

Size: USLetter | Date: | **Rev:**
KiCad E.D.A. 10.0.4 | Id: 19/17



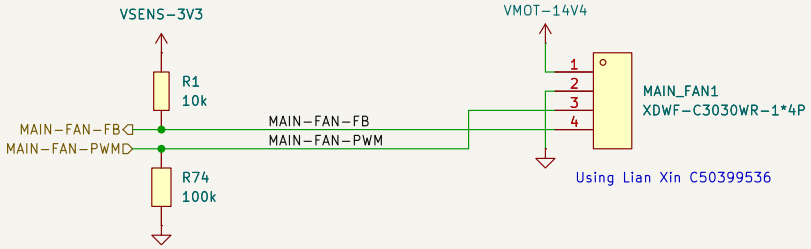
Sheet: /MAIN-BRUSH MOTOR/	
File: MAIN-BRUSH-MOTOR .kicad_sch	
Title:	
Size: USLetter | Date:	**Rev:**
KiCad E.D.A. 10.0.4	Id: 22/17



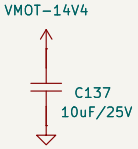
Sheet: /WHEEL MOTOR/
File: WHEEL-MOTORs .kicad_sch

Title:

Size: USLetter | Date: | **Rev:**
KiCad E.D.A. 10.0.4 | Id: 23/17



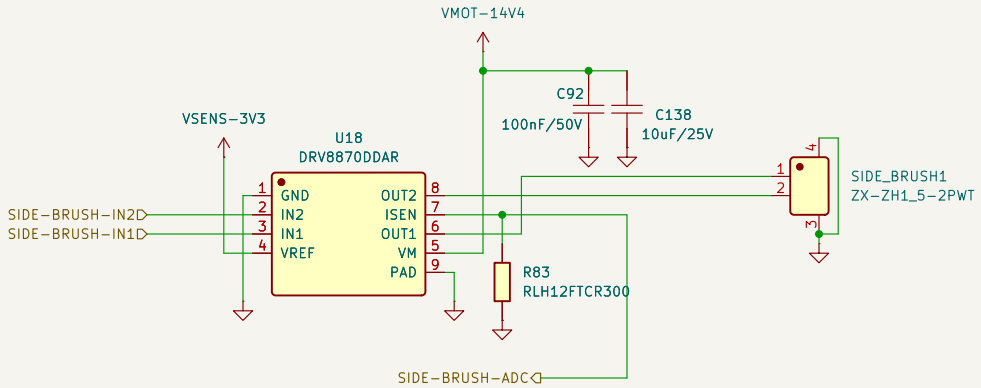
- FB feedback signal is open collector
- set PWM LOW to turn fan OFF
- MAIN_FAN1 matches Dreame L60 PE Ultra PCB



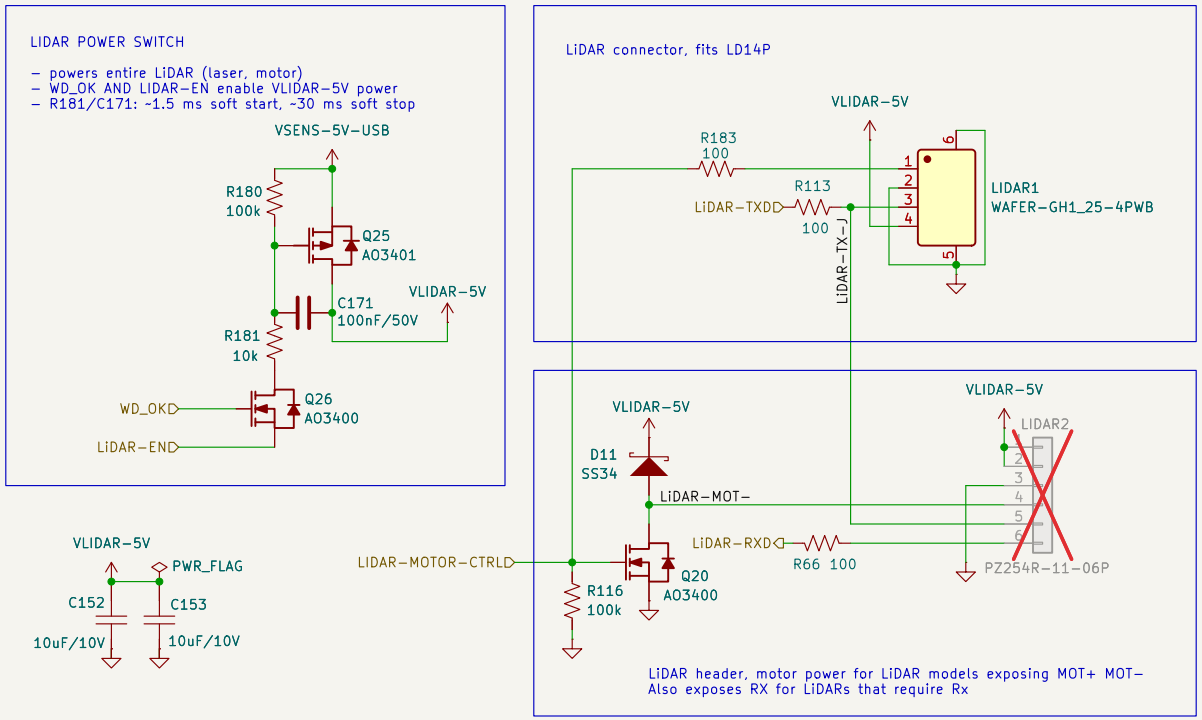
Sheet: /MAIN-FAN/
File: MAIN-FAN .kicad_sch

Title:

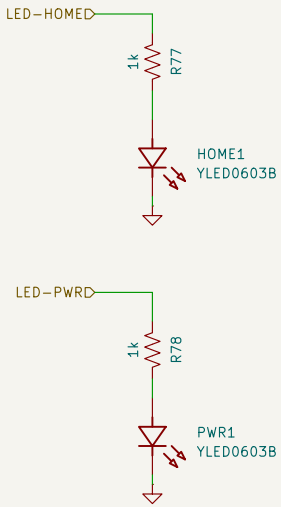
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KiCad E.D.A. 10.0.4		Id: 24/17

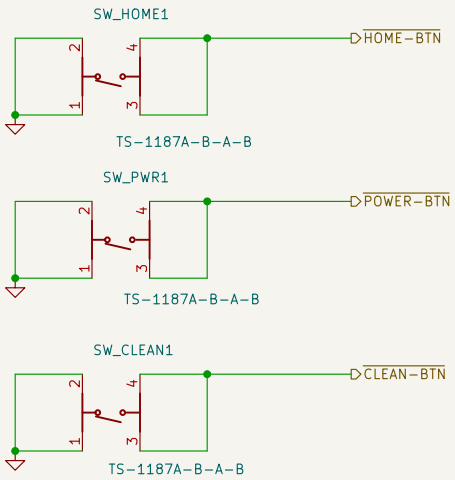


Sheet: /SIDE-BRUSH/		
File: SIDE-BRUSH-MOTORs .kicad_sch		
Title:		
Size: USLetter	Date:	**Rev:**
KiCad E.D.A. 10.0.4		Id: 25/17

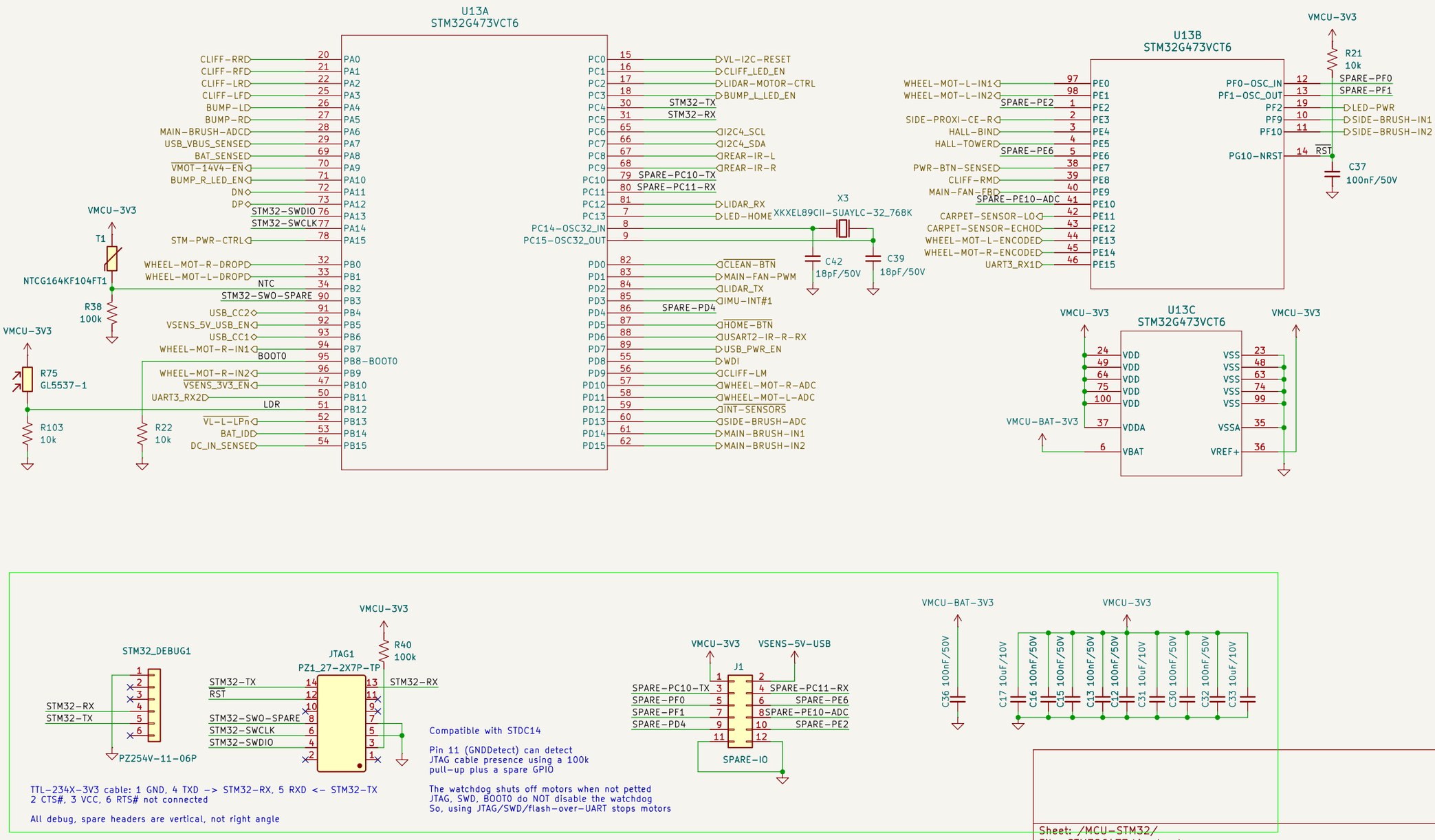


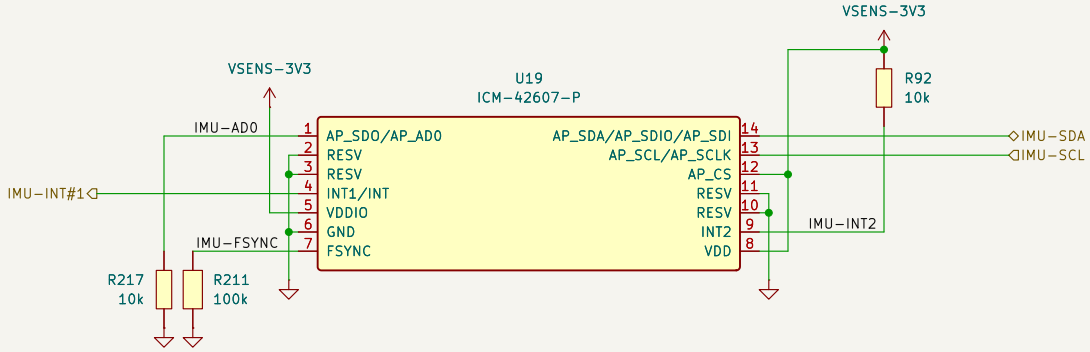
Sheet: /LIDAR/		
File: LIDAR.kicad_sch		
Title:		
Size: USLetter	Date:	**Rev:**
KiCad E.D.A. 10.0.4		Id: 26/17



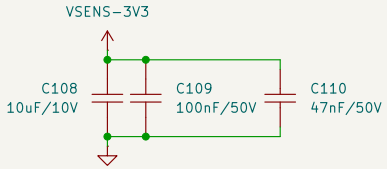


Sheet: /BUTTON-LED/		
File: BUTTON-LEDs .kicad_sch		
Title:		
Size: USLetter	Date:	**Rev:**
KiCad E.D.A. 10.0.4		Id: 28/17


U13A
STM32G473VCT6
CLIFF-RR 20 PA0
CLIFF-RF 21 PA1
CLIFF-LR 22 PA2
CLIFF-LF 25 PA3
BUMP-LD 26 PA4
BUMP-RD 27 PA5
MAIN-BRUSH-ADC 28 PA6
USB_VBUS_SENSED 29 PA7
BAT_SENSED 69 PA8
VMOT-14V4-EN 70 PA9
BUMP_R_LED_EN 71 PA10
DN 72 PA11
DP 73 PA12
STM32-SWDIO 76 PA13
STM32-SWCLK 77 PA14
STM-PWR-CTRL 78 PA15
WHEEL-MOT-R-DROP 32 PB0
WHEEL-MOT-L-DROP 33 PB1
NTC 34 PB2
STM32-SWO-SPARE 90 PB3
USB_CC2 91 PB4
VSENS_5V_USB_EN 92 PB5
USB_CC1 93 PB6
WHEEL-MOT-R-IN1 94 PB7
BOOT0 95 PB8-BOOT0
WHEEL-MOT-R-IN2 96 PB9
VSENS_3V3_EN 47 PB10
UART3_RX2 50 PB11
LDR 51 PB12
VL-L-LPn 52 PB13
BAT_ID 53 PB14
DC_IN_SENSED 54 PB15
PC0 15 VL-I2C-RESET
PC1 16 CLIFF_LED_EN
PC2 17 LIDAR-MOTOR-CTRL
PC3 18 BUMP_L_LED_EN
PC4 30 STM32-TX
PC5 31 STM32-RX
PC6 65 I2C4_SCL
PC7 66 I2C4_SDA
PC8 67 REAR-IR-L
PC9 68 REAR-IR-R
PC10 79 SPARE-PC10-TX
PC11 80 SPARE-PC11-RX
PC12 81 LIDAR_RX
PC13 7 LED-HOME
PC14-OSC32_IN 8
PC15-OSC32_OUT 9
PD0 82 CLEAN-BTN
PD1 83 MAIN-FAN-PWM
PD2 84 LIDAR_TX
PD3 85 IMU-INT#1
PD4 86 SPARE-PD4
PD5 87 HOME-BTN
PD6 88 USART2-IR-R-RX
PD7 89 USB_PWR_EN
PD8 55 WDI
PD9 56 CLIFF-LM
PD10 57 WHEEL-MOT-R-ADC
PD11 58 WHEEL-MOT-L-ADC
PD12 59 INT-SENSORS
PD13 60 SIDE-BRUSH-ADC
PD14 61 MAIN-BRUSH-IN1
PD15 62 MAIN-BRUSH-IN2
X3
XKXEL89CII-SUAYLC-32_768K
C42 18pF/50V
C39 18pF/50V
VMCU-3V3
T1
NTCG164KF104FT1
R38 100k
R75 GL5537-1
R103 10k
R22 10k
U13B
STM32G473VCT6
WHEEL-MOT-L-IN1 97 PE0
WHEEL-MOT-L-IN2 98 PE1
SPARE-PE2 1 PE2
SIDE-PROXI-CE-R 2 PE3
HALL-BIN 3 PE4
HALL-TOWER 4 PE5
SPARE-PE6 5 PE6
PWR-BTN-SENSED 38 PE7
CLIFF-RM 39 PE8
MAIN-FAN-FB 40 PE9
SPARE-PE10-ADC 41 PE10
CARPET-SENSOR-LO 42 PE11
CARPET-SENSOR-ECHO 43 PE12
WHEEL-MOT-L-ENCODER 44 PE13
WHEEL-MOT-R-ENCODER 45 PE14
UART3_RX1 46 PE15
PF0-OSC_IN 12 SPARE-PF0
PF1-OSC_OUT 13 SPARE-PF1
PF2 19 LED-PWR
PF9 10 SIDE-BRUSH-IN1
PF10 11 SIDE-BRUSH-IN2
PG10-NRST 14 RST
R21 10k
C37 100nF/50V
U13C
STM32G473VCT6
24 VDD
49 VDD
64 VDD
75 VDD
100 VDD
37 VDDA
6 VBAT
VMCU-BAT-3V3
23 VSS
48 VSS
63 VSS
74 VSS
99 VSS
35 VSSA
36 VREF+
STM32_DEBUG1
PZ254V-11-06P
JTAG1
PZ1_27-2X7P-TP
R40 100k
STM32-TX
RST
STM32-SWO-SPARE
STM32-SWCLK
STM32-SWDIO
STM32-RX
Compatible with STDC14
Pin 11 (GNDDetect) can detect JTAG cable presence using a 100k pull-up plus a spare GPIO
The watchdog shuts off motors when not petted
JTAG, SWD, BOOT0 do NOT disable the watchdog
So, using JTAG/SWD/flash-over-UART stops motors
TTL-234X-3V3 cable: 1 GND, 4 TXD -> STM32-RX, 5 RXD <- STM32-TX
2 CTS#, 3 VCC, 6 RTS# not connected
All debug, spare headers are vertical, not right angle
J1
VSENS-5V-USB
SPARE-PC10-TX
SPARE-PF0
SPARE-PF1
SPARE-PD4
SPARE-PC11-RX
SPARE-PE6
SPARE-PE10-ADC
SPARE-PE2
SPARE-IO
C36 100nF/50V
C17 10uF/10V
C16 100nF/50V
C15 100nF/50V
C13 100nF/50V
C12 100nF/50V
C31 10uF/10V
C30 100nF/50V
C32 100nF/50V
C33 10uF/10V
Sheet: /MCU-STM32/
File: STM32G473.kicad_sch
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Size: USLegal
Date:
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Rev:
Id: 30/17



IMU on I2C4 (2.2k pull-ups R5/R6 on the top sheet). AP_AD0 pulled low by R217 (I2C address 0x68).
The IMU is leadless: probe AD0 on R217, INT2 on its pull-up R92 and FSYNC on R211. Run a wire to a spare pin if INT2 or FSYNC is needed.

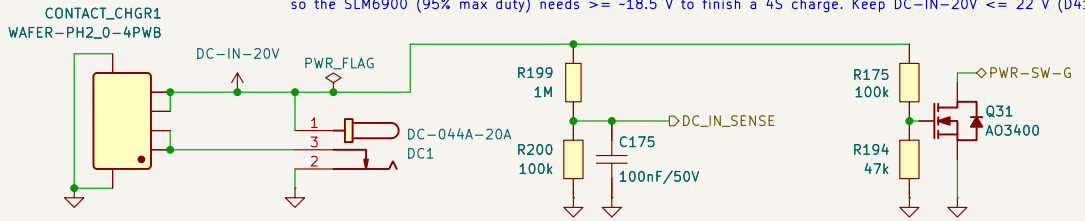


Sheet: /IMU/
File: IMU-ICM-4267-P .kicad_sch

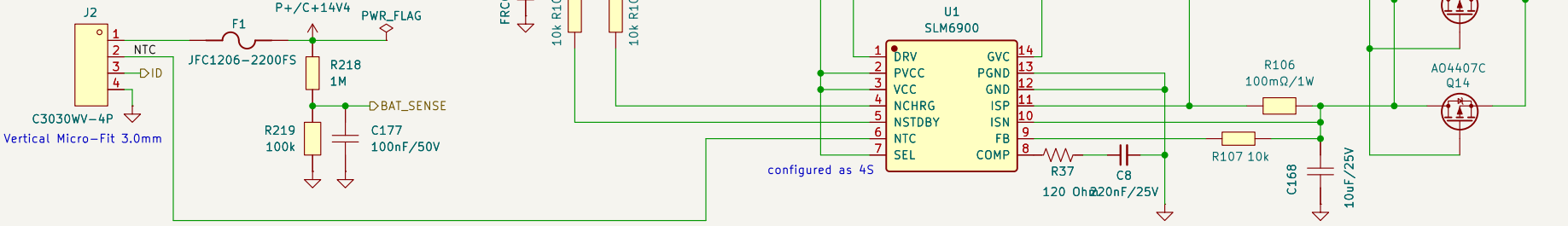
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DC−IN−20V: 20 V brick or dock. With the battery disconnected (Q14 off) VBAT−14V4 = DC−IN−20V − 0.4 V, so the SLM6900 (95% max duty) needs >= ~18.5 V to finish a 4S charge. Keep DC−IN−20V <= 22 V (D41, Q13 Vgs).



MAIN BATTERY CONNECTOR



Rail−hold: charger power (dock or barrel jack) turns the MCU rail on and keeps it on. DC−IN−20V to GND ~130k (R175+R194 || R199+R200) = dock presence signature.

R106 sets charging current 100 mOhm 1.2 A

Sheet: /BATTERY−CHARGER/	
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